



Git and GitHUB





What is Git?

Git is a distributed version control system (VCS).

Git is a tool that helps multiple people work on the same code project or documents by tracking and managing changes to the files.



Project Files





Project Files

Bug Fix

New Updates

Security Updates



Project Files

Bug Fix



New Updates



Security Updates





What is VCS?



Every time you make a change, whether it's adding a sentence to a document or altering a line of code, the VCS records and saves the outcome.



project



project-1



project-2



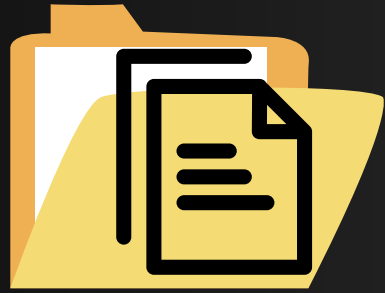
project-final



project-final1



project-finalfinal



Project Files



Project

1

Project

Project



Project

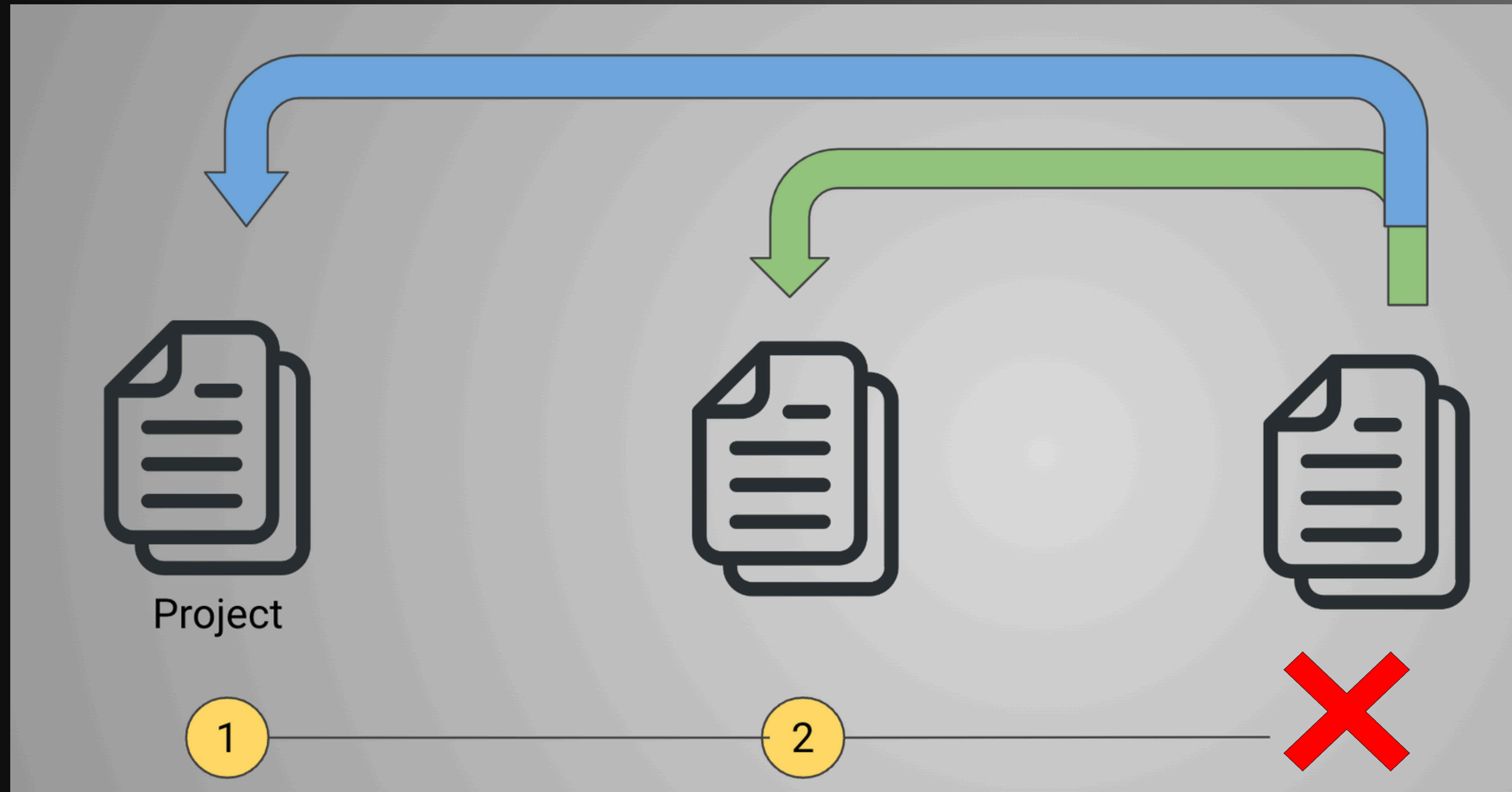
1



Project

2





Why VCS?



- **Backup and Restore:** Files are safe against accidental losses or mistakes.
- **Collaboration:** Multiple people can work on the same project simultaneously.
- **Branching and Merging:** Users can diverge from the main base of code, experiment, and then bring changes back in line without losing work.
- **Tracking Changes:** You can see specific changes made and by whom.

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Installing Git



- **Step-by-step installation of Git.**
- **Configuring Git with your username and email.**



Configure Git with Username and Email

```
$ git config --global user.name "Paul Philips"
```

```
$ git config --global user.email paulphilips@email.com
```

```
$ git config --list
```



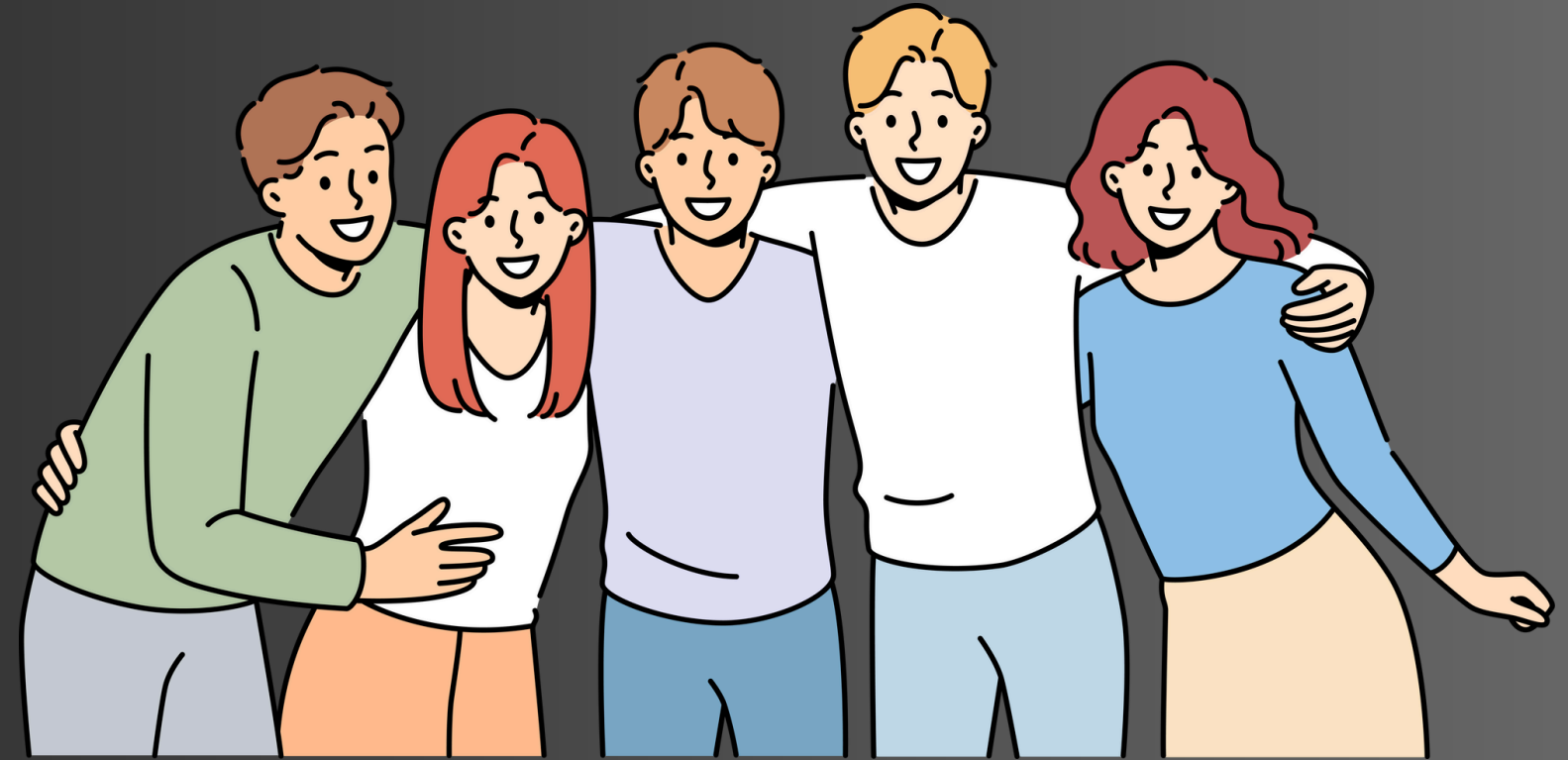
Creating Local Git Repo

```
$ git init
```




What is Git Commit?

A git commit is a command in Git that captures a snapshot of the project's currently staged changes, creating a permanent record in the repository's history.





Working
Directory

Staging

Git Local Repo

File

Git add

Git commit

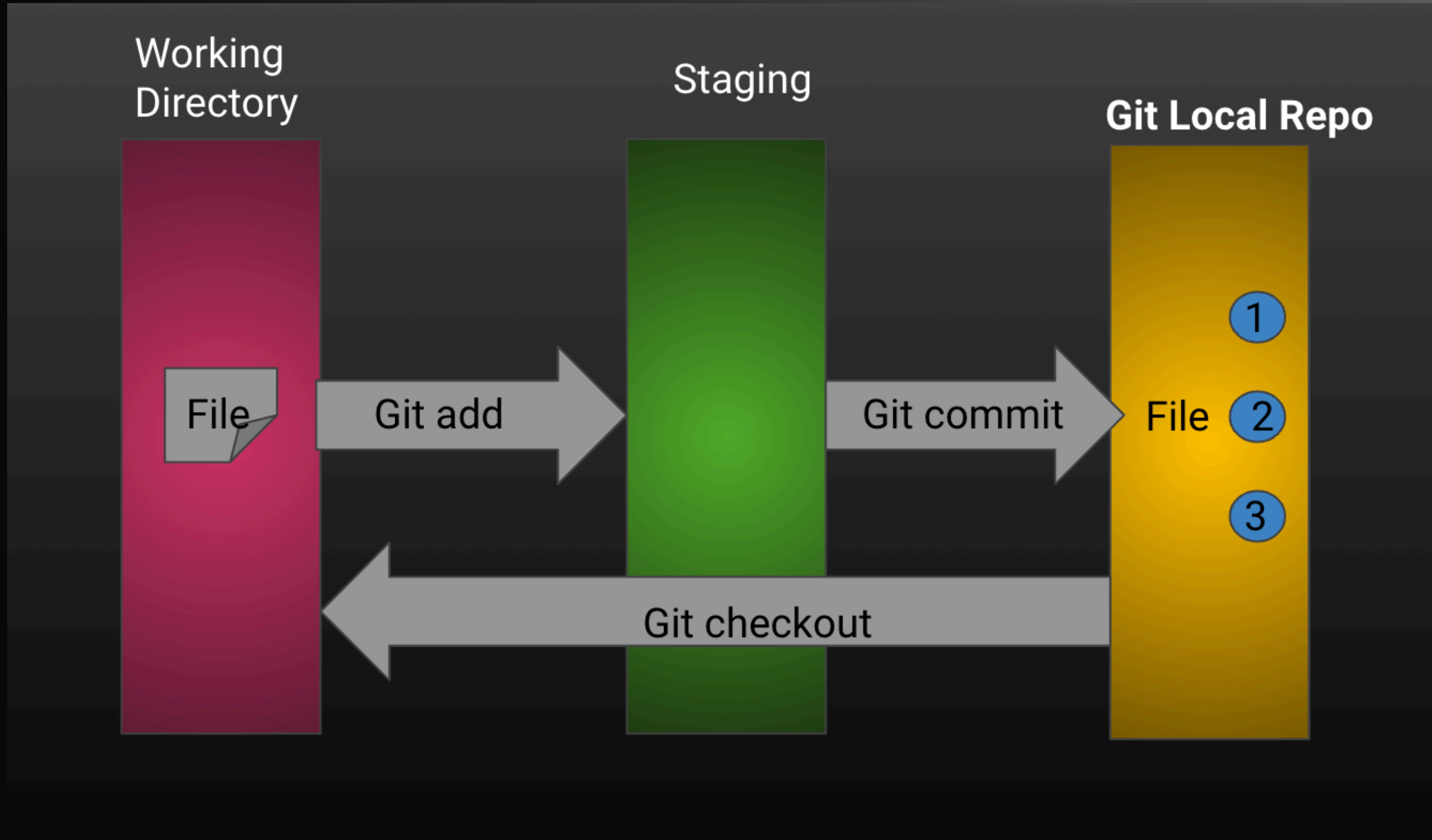
File

1

2

3

Git checkout



Negative Cases



If we made any change by mistake and save it

Case1: To undo changes, get the last successful change

`git restore . or filename (. mean all files)`

Case2: If we added the changes using git add then..

`git restore --staged <file_path> # To unstage`

`git restore <file_path> # To discard changes in the working directory`

Negative Cases



Case3: Added changes to staging area (didn't commit) after this added more changes to file

//To get the staged changes

`git restore --worktree index.html`

Case4: How about if we did commit also wrong files

`git reset --soft HEAD^` (uncommit and keep the changes)

`git reset --hard HEAD^` (uncommit and discard the changes)

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Logging in Git



Useful Log Options

- `git log -p -2` (*last two commit with diff*)
- `git log --stat` (*summary of changes*)
- `git log --pretty=oneline`
- `git log --pretty=format:"%h - %an, %ar : %s"`
- `git log -S function_name`



Useful Log Options

- `git log --grep="fix bug"` (*search commit msg*)
- `git log --since="2024-01-01"`
- `git log --until="2024-01-01"`
- `git log --author="Paul"`
- `git log --no-merges`

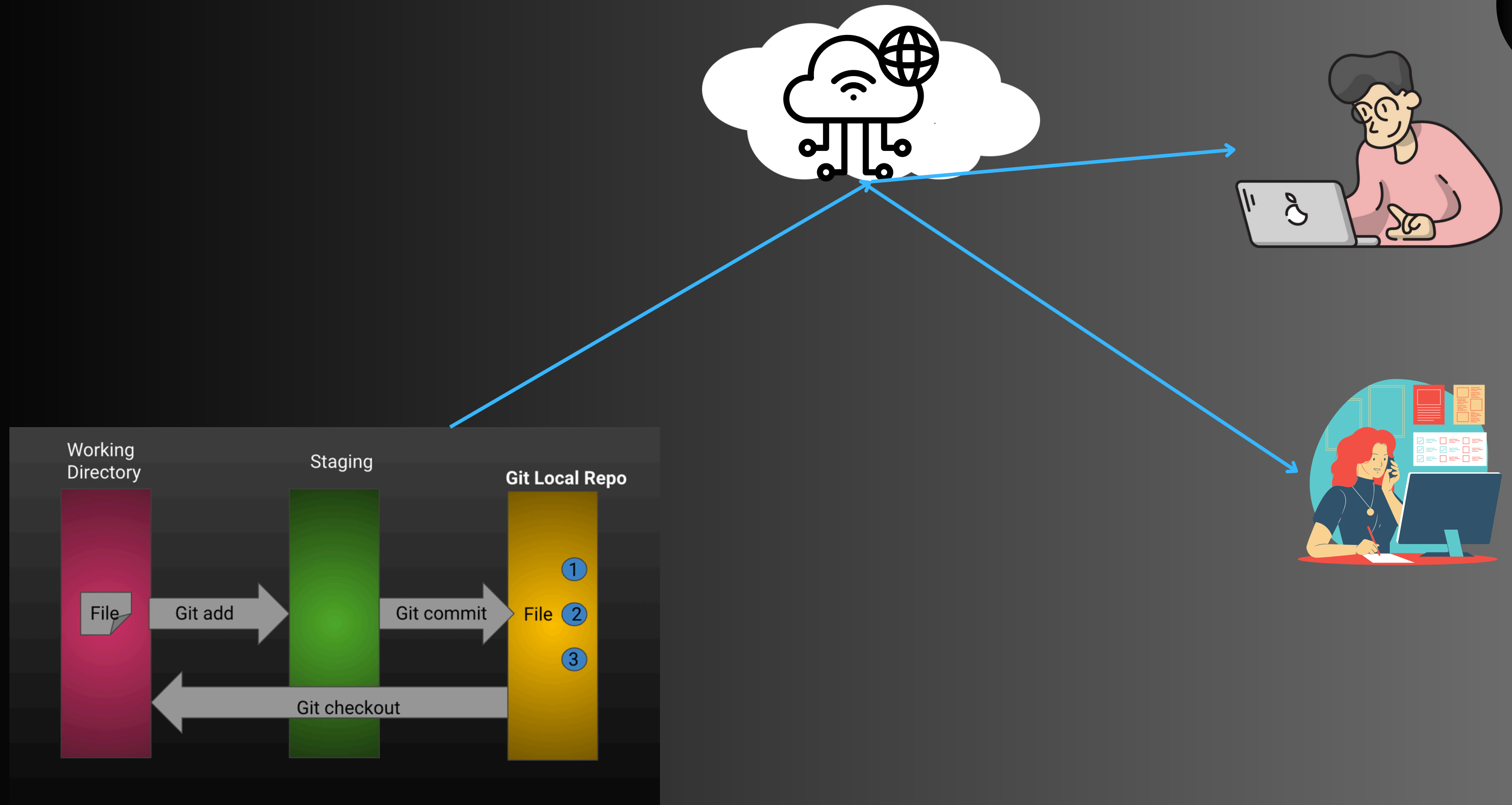
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Remote Repo



A remote repository refers to a version of your project that resides on a network server or a hosted repository on the internet.





Git HUB





Adding Remote Repo

```
git remote add origin <remote URL>
```

```
git push -u origin master
```




Working with Remote

To see the remote git link

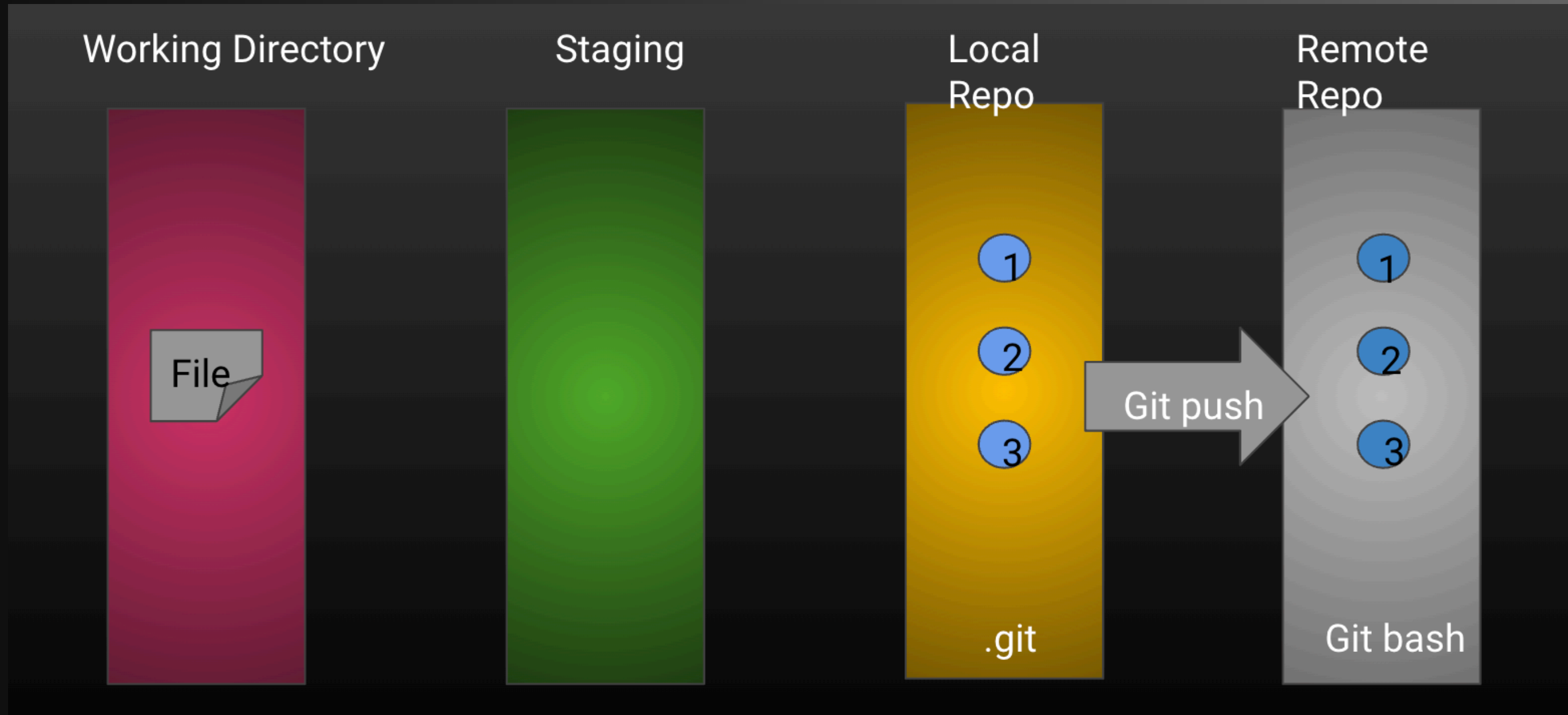
`git remote`

`git remote -v`

`git remote show origin (to get more info about remote)`

`git clone`

`git pull`





Git status	To check for any change in the repo
Git diff	To see the difference of current version with last committed version
Git pull	To fetch the latest files from remote repo
Git add	To add the modified files in staging area
Git commit	To add the changes in git repo
Git push	To push the committed changes in Remote repo
Git log	To see the old commits history



Branching and Merging

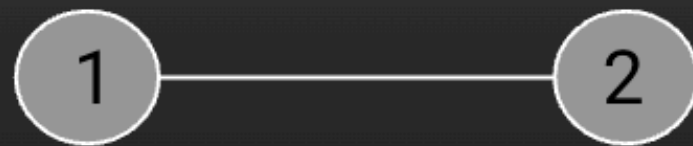
master branch





Branching and Merging

master branch



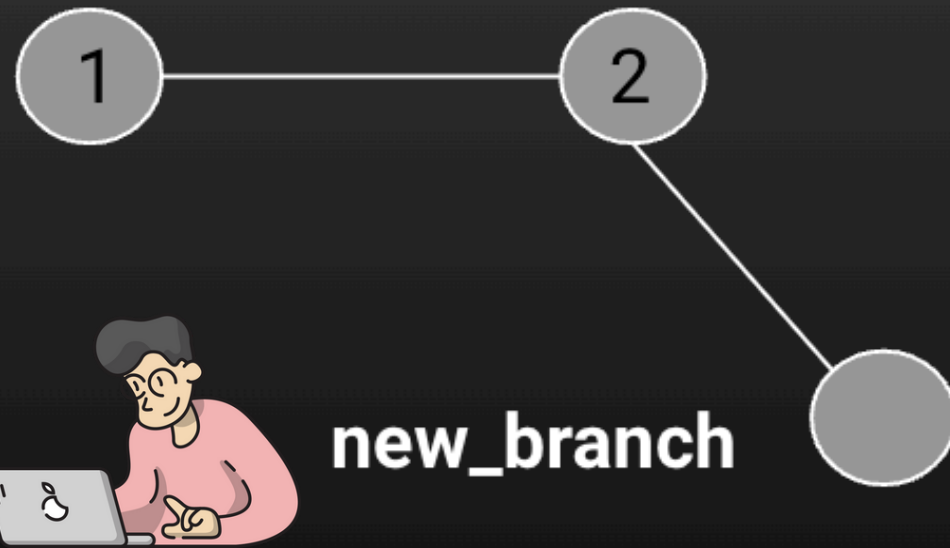
I can add
design





Branching and Merging

master branch

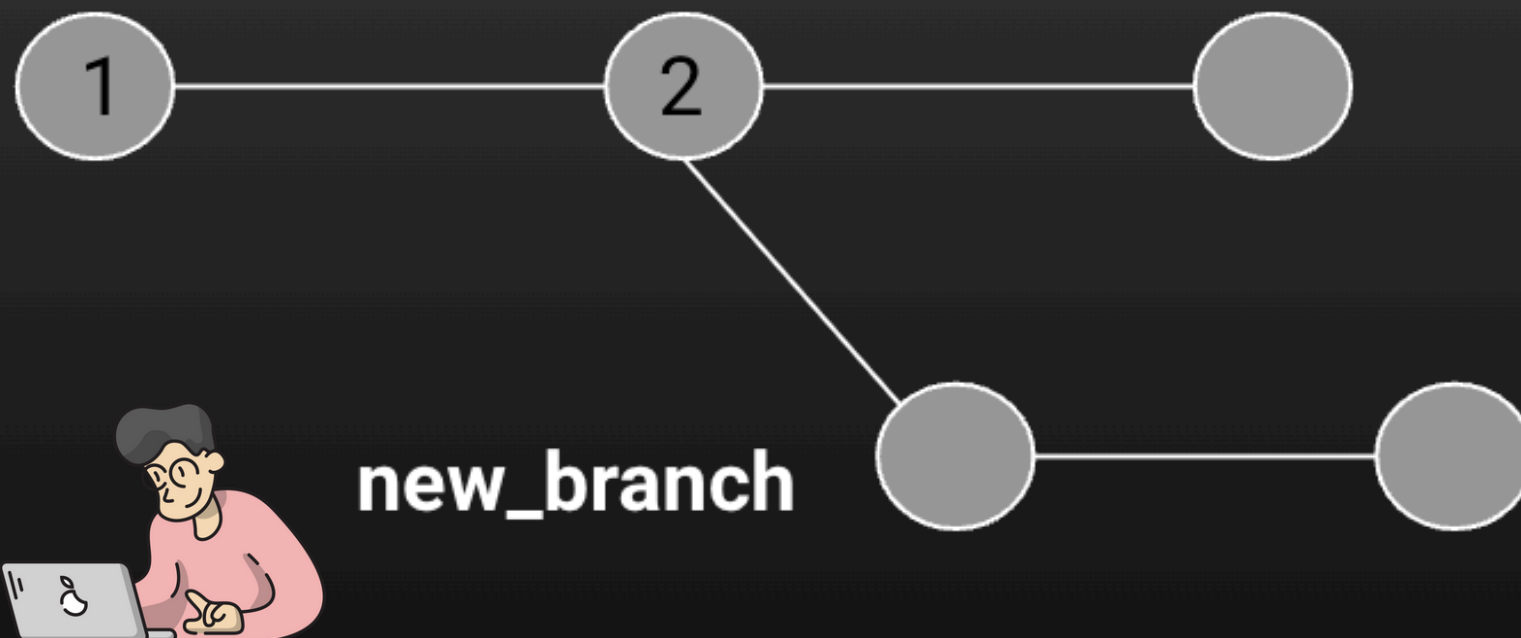


```
$ git branch new_branch
```



Branching and Merging

master branch



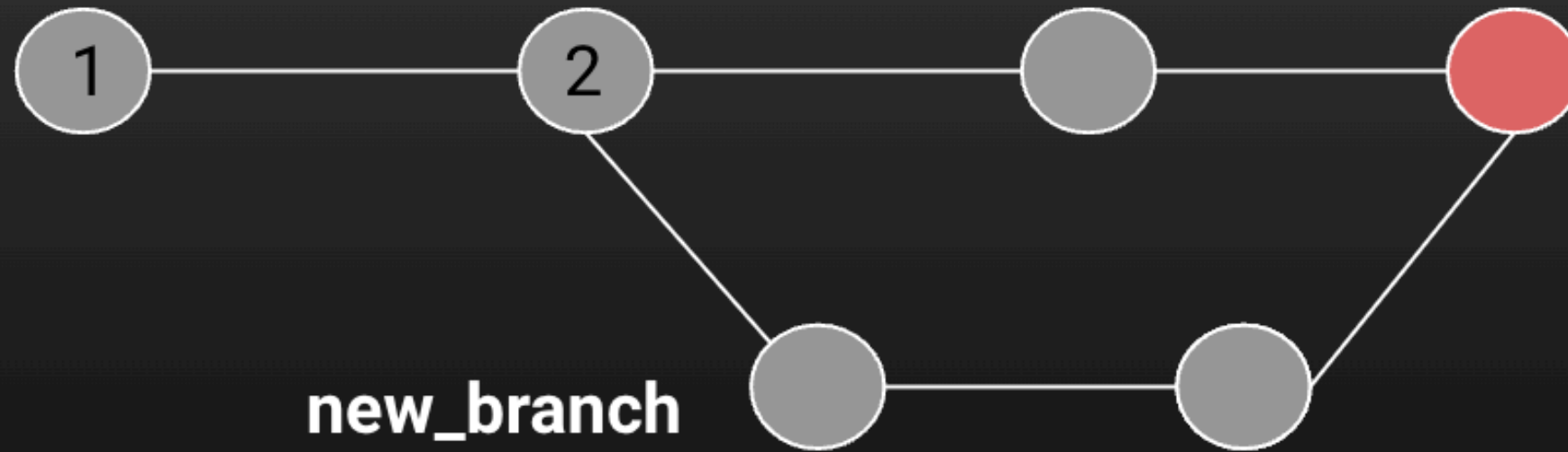
new_branch

```
$ git branch new_branch
```



Branching and Merging

master branch

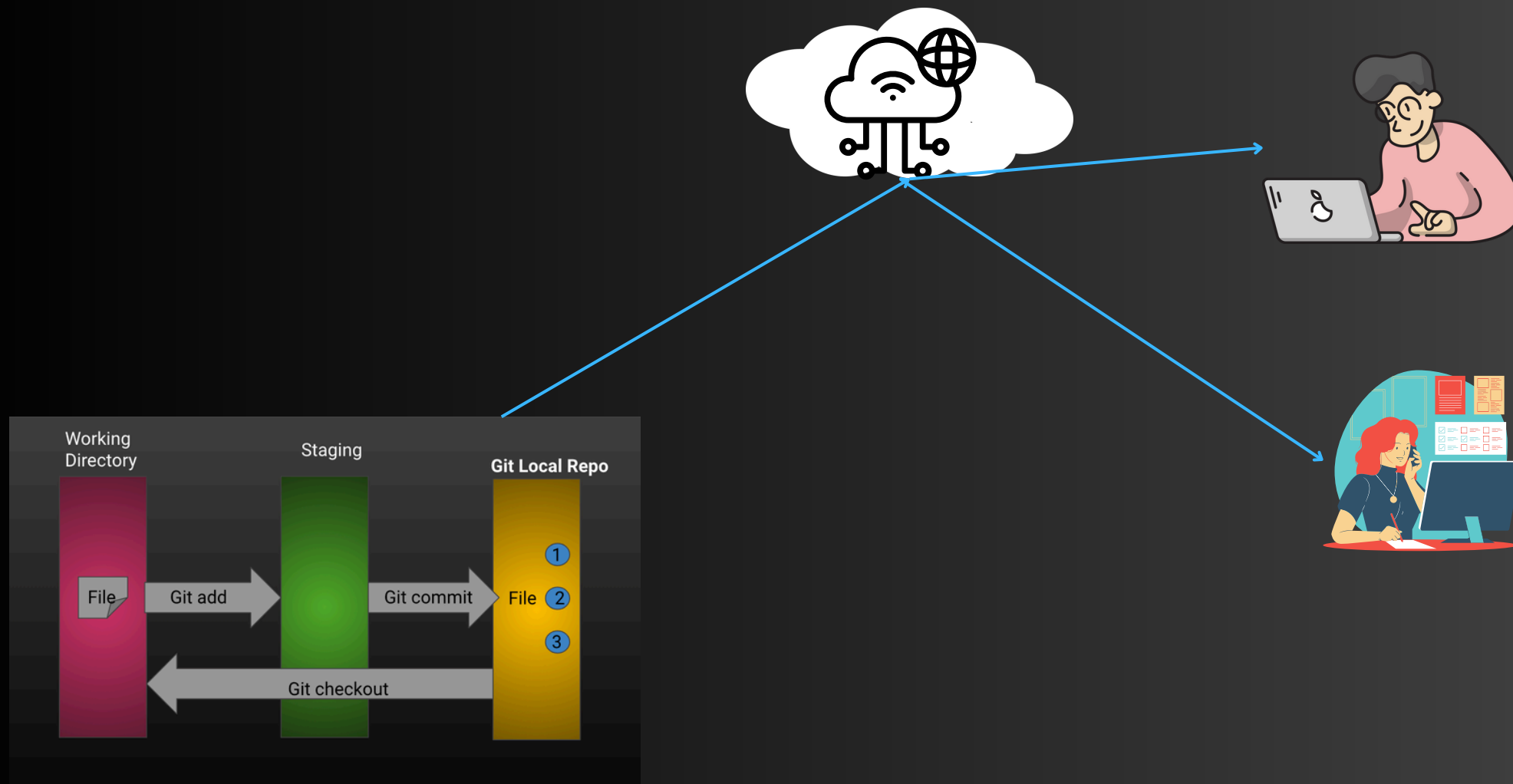


```
$ git checkout master  
$ git merge new_branch
```


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Git Forking and Pull Request



I can improve
the code



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Gitignore



Git Ignore

In case you don't want to commit some files from your working directories then add those files name in **.gitignore** file.

Will be useful when your file contain confidential info.



Cloning

To get the remote repo into your local system.

```
git clone <remote_repo_link>
```

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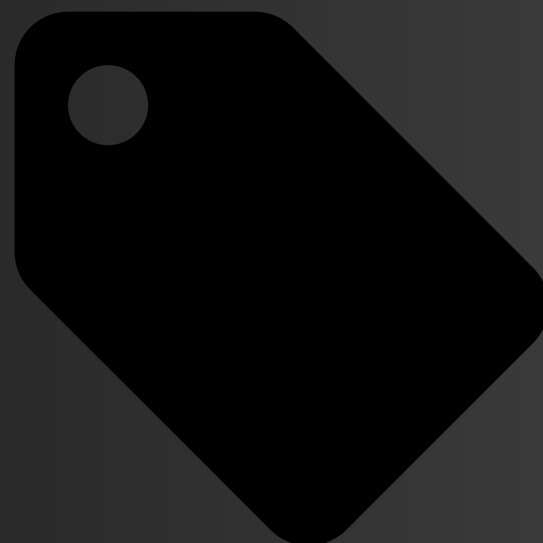


Git Clean

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Git Tags





Git Tagging

To create annotated tags

```
git tag -a v1.0 -m "My version 1.0"
```

To show tags data

```
git show v1.0
```

To tag old commit in case you forgot

```
git tag -a v1.2 <commit_no>
```




Git Tagging

To delete a tag

```
git tag -d <tag_no>
```

Tags created remain local, to move it to remote:

```
git push origin v1.5
```

```
git push origin --tags (For all tags together)
```

In case in future, if you wanna give patch for v1.2 release let's say

```
git checkout -b version2 v2.0.0 (it will create new branch also)
```



Managing access and permissions.

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Using GitHub for Project Management



- **Overview of GitHub features like Issues, GitHub Actions, and Projects.**
- **Tracking project progress using Kanban boards.**
- **Automating workflows with GitHub Actions.**



- **Securing your Git and GitHub accounts.**
- **Managing access and permissions.**
- **Monitoring repositories for vulnerabilities.**